

Modernization of a **BIBO 3D Printer System**

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Live Print Demo

Video of the upgraded printer in action



The Starting Point

What we received — and what was wrong with it

BIBO 2 Touch — Stock Specs

Kinematics:	Cartesian (bed Z, head X/Y)
Extruder:	Dual Bowden-style (×2)
Firmware:	Stock Klipper (without configuration)
Bed:	Aluminium base + glass sheet
Frame:	Aluminium extrusion + acrylic
Status at start:	NON-FUNCTIONAL

Problems Identified

- ✗ Printer was completely non-functional
- ✗ Dual extruder added mass & limited print size
- ✗ Glass bed was unstable and hard to level, and difficult to remove printed parts
- ✗ Front door had no latch
- ✗ Poor internal lighting — hard to monitor prints
- ✗ Noise and vibration while printing



Hardware Upgrades

Six targeted improvements — each solving a specific problem



Single Extruder

✓ Complete

Reduced carriage mass, removed nozzle offset errors, gained full bed travel



PEI Magnetic Bed

✓ Complete

Replaced unstable glass sheet — reliable leveling, easy part removal by flexing



Motherboard(SKR Mini E3 V3.0)

✓ Complete

interpolates the stepping signals to eliminate the high-pitched electrical whining and resonant mechanical noise typically generated by standard motor drivers during axes movement



Metal Dual-Gear Extruder

✓ Complete

Replaced plastic single-gear — stronger filament grip, eliminates slipping



Internal LED Lighting

✓ Complete

12V LED strip along top frame — fully visible print area during operation



Vibration Damping Legs

✓ Complete

TPU 95A printed feet — absorb printer vibration, reduce ringing artifacts

Firmware Migration

From legacy Marlin on a microcontroller → Klipper running on Linux

BEFORE — Marlin

- Runs entirely on 8-bit MCU
- Any config change = recompile & reflash
- No web interface
- No input shaping
- No pressure advance
- Limited to MCU compute power



AFTER — Klipper

- ✓ Host runs on Linux workstation
- ✓ Config = edit text file, restart
- ✓ Mainsail web UI over local network
- ✓ Input shaping via ADXL345
- ✓ Pressure advance support
- ✓ Moonraker REST/WebSocket API

Stack: KIAUH → Klipper → Moonraker → Mainsail



Problems We Solved — Hardware

Real issues encountered, diagnosed, and fixed during the project

P1

Extruder Fan Blocked X-Axis Travel

Symptom: Active nozzle couldn't reach the far end of the bed

Fix:

Removed the cooling fan from the disabled extruder; re-configured axis endpoint in firmware

P2

Filament Jam in Hot End

Symptom: Filament stuck in nozzle — couldn't feed or retract

Fix:

Standard unclog failed; physically replaced the nozzle/extruder tip on the jammed unit

P3

New Metal Extruder Too Large to Fit

Symptom: Dual-gear extruder didn't fit the existing carriage bracket

Fix:

Designed and 3D-printed two new custom parts: a filament-path adapter and a full motor/fan/extruder mount

P4

Printed parts undersized/under extrusion

Symptom: Printed parts were consistently smaller than designed. A test square designed to be 20 mm measured only 16.5 mm

Fix:

The correct esteps was calculated for both the axis and the extruder rotational distance



Problems We Solved — Firmware

Configuration challenges during Klipper setup

P5

Moonraker Couldn't Connect to Klipper

Fix:

Socket path mismatch: default was `/home/pi/...` but our user was `cpsstudent`. Corrected `klippy_uds_address` in `moonraker.conf`.

P6

All Endstops Permanently Showing TRIGGERED

Fix:

The `^` pull-up prefix in `printer.cfg` inverted logic on NC endstops. Removed `^` from all three axis pin definitions.

P7

MCU Firmware Version Mismatch

Fix:

Host was v0.13, MCU was v0.12. Recompiled with `make menuconfig` (ATmega2560, 250000 baud) and reflashed.

P8

Printed Parts Consistently Undersized (16.5mm vs 20mm)

Fix:

Wrong `rotation_distance`. Recalculated: $(16.5/20) \times 40 = 33$. Updated in `printer.cfg` — confirmed with reprint.

P9

Under-Extrusion (Weak Layers, Gaps)

Fix:

PID tuning via `PID_CALIBRATE` command at 200°C (hotend) and 60°C (bed). E-steps also recalibrated by measuring extruded filament.



Performance Testing

Benchmark: 3DBenchy tugboat — same G-code, same parameters, before and after

Why 3DBenchy?

Industry-standard FDM benchmark · Tests overhangs, bridging, curved surfaces, small details, and dimensional accuracy · Same G-code file used pre- and post-upgrade to ensure fair comparison

BEFORE UPGRADES

Print time:	~1h 24min (estimated baseline)
Dimensional accuracy:	Parts consistently undersized
Surface quality:	Under-extrusion gaps, weak layers
Bed adhesion:	Glass sheet — unreliable, parts stuck
Vibration artifacts:	Ringing visible on curved surfaces



AFTER UPGRADES

Print time:	✓ 1h 19min (actual — Mainsail log)
Dimensional accuracy:	✓ Correct after rotation_distance fix
Surface quality:	✓ Solid layers, no extrusion gaps
Bed adhesion:	✓ PEI sheet — reliable, pops off cleanly
Vibration artifacts:	✓ Reduced by damping legs + lighter head

Total filament used: 4.44 m · Slicer estimate: 1:08:14 · Actual print time: 1:19:20 · Total session: 1:24:21



Live Print Demo

Upgraded BIBO printing the 3DBenchy benchmark



[INSERT VIDEO HERE]

Insert > Video (PowerPoint) — or drag and drop your .mp4 file onto this slide

 Trim to key moments: · first layer, mid-print, part removal

 Note: printer should be · quieter with damping legs

 LED lighting makes · print area clearly visible

Summary & What's Next

What We Achieved

- ✓ Restored a non-functional printer to full operation
- ✓ Migrated to Klipper with Mainsail web interface
- ✓ Removed dual extruder → lighter, simpler system
- ✓ PEI magnetic bed → reliable adhesion & easy removal
- ✓ Metal dual-gear extruder → no more slipping
- ✓ Thermal insulation → faster heat-up, stable temperature
- ✓ LED lighting → full visibility of print area
- ✓ TPU damping legs → reduced vibration & ringing
- ✓ Diagnosed and fixed 14 documented problems

Future Work

- ADXL345 input shaping
- Automated nozzle purge bucket
- 5015 blower for better cooling
- Adaptive speed printing
- Camera-based failure detection
- Predictive maintenance via vibration data
- Professional cable management

Thank You — Questions?